

1. Consider a spherical shell of inner and outer radius a and b , respectively, and conducting a uniform current density $\vec{J} = \hat{\phi} J_0$, (where J_0 is a constant), which is always tangent to circles of constant latitude, as shown in Fig. 1. Find the vector potential $\vec{A}$ and the $\vec{B}$ field at the centre of the sphere (i.e. at $z=0$)

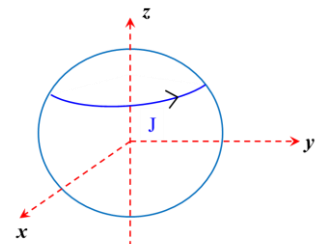
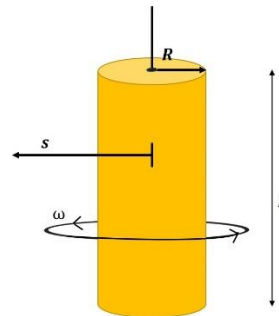


Fig.1

2. An infinitely long conductor of radius a carries a uniform current with $\vec{J} = J_0 \hat{z}$. Find the magnetic vector potential for $\rho < a$.
3. A uniformly charged disk of radius R carrying a charge Q is rotating with angular velocity ω about its axis. Find its magnetic moment.
4. A thin glass rod of radius R and length L carries a uniform surface charge σ . It is set spinning about its axis, at an angular velocity ω . Find the magnetic field, $\vec{B}$ at a distance $s \gg R$ from the axis, in the $x-y$ plane (see below Fig.2). Clue: consider it as a stack of magnetic dipoles.



5. A flat circular disc of radius a has a uniform surface charge density $-\sigma$ from its centre up to a certain radius r and a uniform surface charge density $+\sigma$ from r up to a . The total charge on the disc is zero. The mass of the disc is M and it is distributed uniformly over its area. If the disc is rotating with angular velocity $\vec{\omega}$ about its figure axis, calculate the total magnetic moment. Also find the gyromagnetic ratio.
6. Determine the torque and force between two circular conductors with parallel axes carrying equal and equidirectional currents I if the distance between the centres of the loops is large ($L \gg R$). Express the force and the torque as a function of the angle θ between the tie line and the conductor plane (see Fig.2 below).

